

# Customer Shopping Behavior Analysis Project

## Python + SQL + Power BI | End-to-End Data Analytics Project

### Project Overview

This project demonstrates a complete end-to-end Data Analytics workflow using **Python**, **SQL**, and **Power BI** to analyze customer shopping behavior and business performance. The objective was to transform raw retail transaction data into meaningful business insights that support data-driven decision making.

The project begins with data preprocessing and preparation in **Python**, where the dataset is cleaned, transformed, and enriched by creating additional business-related features. The processed data is then stored in a **MySQL database**, where SQL is used to perform exploratory analysis, calculate key business metrics, and answer real-world business questions. Finally, the analyzed data is visualized through interactive **Power BI dashboards**, enabling stakeholders to monitor sales performance, customer behavior, product trends, and business growth.

The dashboard is designed for executives, business managers, marketing teams, and sales analysts to quickly understand customer purchasing patterns and identify opportunities for improving revenue and customer satisfaction.

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### Project Workflow

#### 1. Data Collection

- Retail customer shopping dataset
- Customer demographics
- Product information
- Purchase history
- Ratings
- Shipping details
- Subscription status

#### 2. Data Processing (Python)

Python was used to prepare the dataset before loading it into SQL.

Tasks performed include:

- Data Cleaning
- Handling Missing Values
- Removing Duplicate Records
- Feature Engineering
- Creating Customer Segments
- Creating Age Groups
- Purchase Frequency Analysis

- Exporting Clean Dataset

### 3. Database Analysis (SQL)

The cleaned dataset was imported into MySQL where SQL queries were written to answer business questions.

Analysis included:

- Total Revenue
- Average Purchase Value
- Customer Count
- Revenue by Category
- Revenue by Gender
- Revenue by Season
- Revenue by Location
- Customer Loyalty Analysis
- Subscription Analysis
- Discount Analysis
- Product Performance
- Customer Rating Analysis
- Shipping Performance
- Customer Purchase Frequency
- Top Selling Products

The SQL layer converts raw transactional data into business-ready metrics used in Power BI.

### 4. Data Visualization (Power BI)

The Power BI report consists of three interactive dashboards that provide insights at different business levels.

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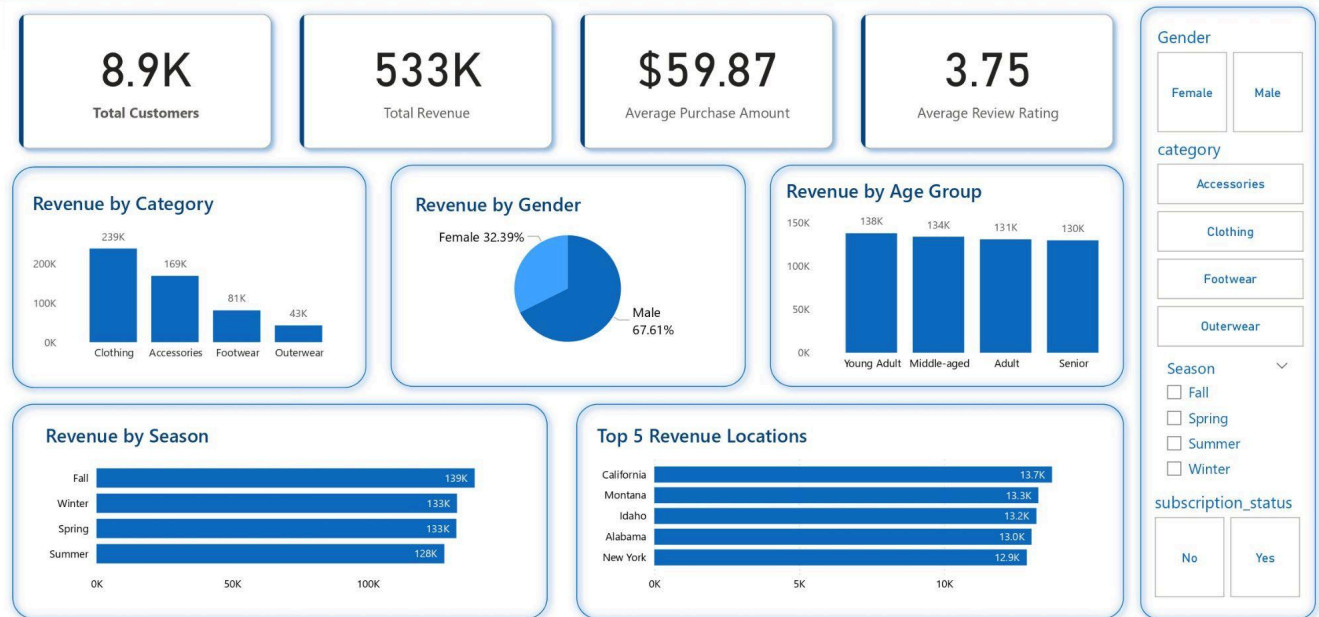
## Dashboard 1

### Executive Sales Overview

#### Purpose

Provides a high-level overview of business performance through KPIs and sales trends.

## Executive Sales Overview



### Key Performance Indicators

- Total Customers: **8.9K**
- Total Revenue: **533K**
- Average Purchase Amount: **\$59.87**
- Average Customer Rating: **3.75**

### Business Insights

Revenue analysis reveals that **Clothing** is the highest revenue-generating category, followed by **Accessories**, while **Outerwear** contributes the least. The customer base is predominantly **male (67.61%)**, indicating that a significant portion of sales originates from male shoppers.

Among different age groups, **Young Adults** generate the highest revenue, followed closely by **Middle-aged** customers, making these demographics the primary revenue contributors.

Seasonal analysis shows relatively consistent sales throughout the year, with **Fall** producing the highest revenue, suggesting stronger customer demand during this season.

Geographical analysis identifies **California, Montana, Idaho, Alabama, and New York** as the top-performing locations, enabling businesses to prioritize marketing campaigns and inventory planning in these regions.

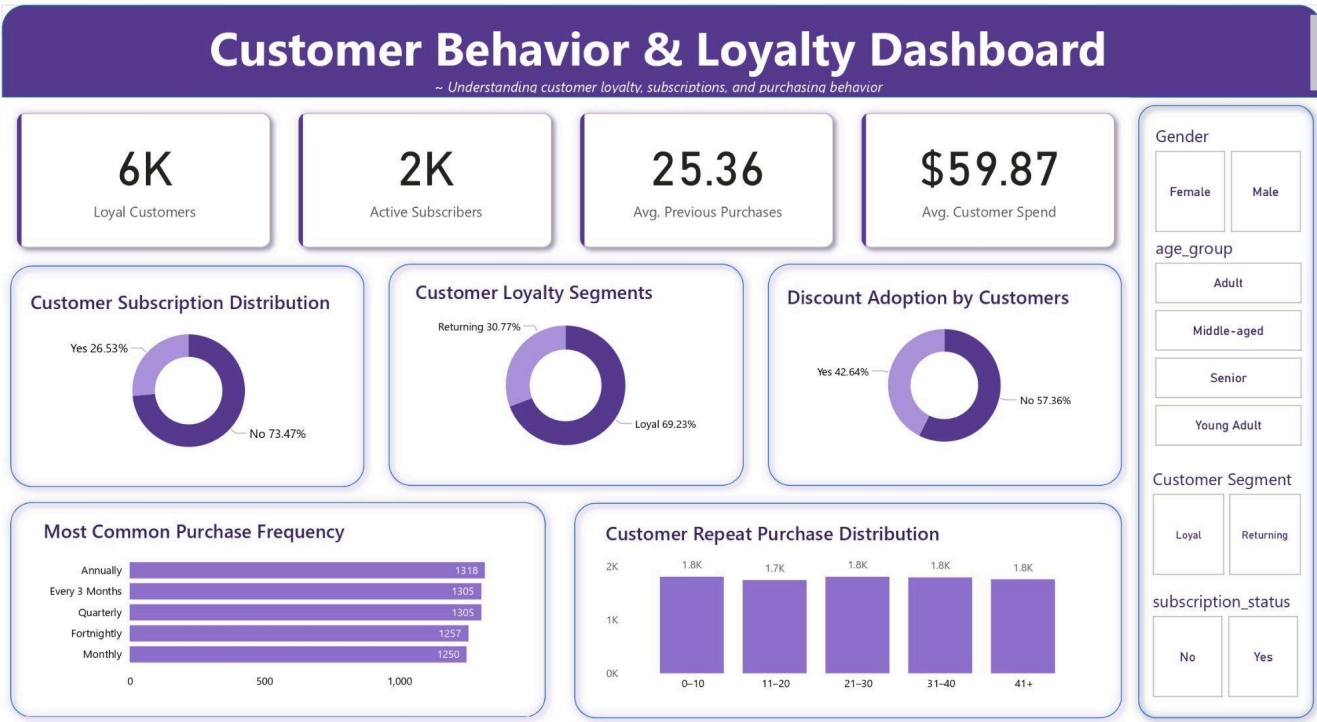
Interactive slicers allow business users to analyze revenue by gender, category, season, and subscription status.

# Dashboard 2

## Customer Behavior & Loyalty Dashboard

### Purpose

Analyzes customer engagement, purchasing habits, and loyalty to understand long-term customer value.



Purchase frequency analysis highlights that customers commonly shop **monthly, quarterly, and annually**, providing useful information for planning promotional campaigns and customer engagement strategies.

Repeat purchase distribution indicates that customers with higher previous purchase counts consistently contribute to revenue, emphasizing the importance of retaining existing customers rather than focusing solely on acquiring new ones.

Interactive filters allow analysis by age group, customer segment, gender, and subscription status.

## Dashboard 3

### Product Performance & Customer Satisfaction

#### Purpose

Evaluates product success, customer satisfaction, and shipping performance to improve business operations.



#### Key Performance Indicators

- Highest Revenue Category: **Clothing**
- Best Selling Product: **Blouse**
- Highest Rated Product: **Gloves**
- Best Shipping Experience: **Standard Shipping**

## Business Insights

Product-level analysis shows that **Blouse**, **Shirt**, **Jewelry**, **T-Shirt**, and **Dress** generate the highest revenue, indicating strong customer demand for these products.

Customer ratings reveal that products such as **Gloves**, **Skirts**, and **Sandals** receive the highest satisfaction scores, suggesting consistent product quality and positive customer experiences.

Shipping analysis demonstrates that **Standard Shipping** achieves the highest average customer rating, while other shipping methods maintain relatively similar satisfaction levels. This suggests that customers value reliability over premium shipping options.

The treemap visualization illustrates category-wise revenue contribution, showing that Clothing dominates overall sales while Accessories and Footwear also contribute significantly.

Interactive slicers enable detailed analysis by category, customer segment, gender, and discount status.

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## Key Business Findings

- Generated total revenue of approximately **\$533K** from **8.9K** customer transactions.
  - Clothing is the highest-performing product category.
  - Blouse is the best-selling product.
  - Young Adult customers contribute the largest share of revenue.
  - Male customers account for nearly two-thirds of total sales.
  - Customer spending remains consistent with an average purchase value of **\$59.87**.
  - Around **69%** of customers are loyal, indicating strong retention.
  - Only about **26%** of customers are subscribed, highlighting potential for subscription growth.
  - Standard Shipping delivers the highest customer satisfaction.
  - Revenue remains stable across all seasons, with Fall showing the strongest performance.
  - California and other top-performing states contribute significantly to total revenue.
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## Business Recommendations

- Increase investment in high-performing product categories such as Clothing and Accessories.
- Develop targeted marketing campaigns for Young Adult customers, who generate the highest revenue.
- Expand subscription programs with exclusive benefits to convert more non-subscribers into loyal members.
- Focus promotional efforts in high-performing locations while exploring growth opportunities in lower-performing regions.
- Continue maintaining product quality for top-rated products and apply similar quality standards to lower-rated items.
- Promote repeat purchases through personalized offers, loyalty rewards, and customer retention initiatives.
- Optimize inventory planning for Fall, when customer demand reaches its peak.
- Use customer segmentation to deliver personalized marketing and improve overall customer lifetime value.

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## Technologies Used

**Programming Language:** Python

**Database:** MySQL

**Query Language:** SQL

**Data Visualization:** Power BI

**Python Libraries:** Pandas | NumPy | SQLAlchemy | PyMySQL

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## Conclusion

This project demonstrates a complete business intelligence solution by integrating **Python, SQL, and Power BI** into a single analytics pipeline. Python was used for data preparation and feature engineering, SQL transformed transactional data into meaningful business metrics, and Power BI presented those insights through interactive dashboards. The analysis highlights customer purchasing behavior, sales performance, product trends, loyalty, subscription adoption, and customer satisfaction, enabling stakeholders to make informed business decisions. Overall, the project showcases practical skills in data cleaning, SQL querying, KPI development, dashboard design, and business storytelling, making it a strong portfolio project for a Data Analyst or Business Intelligence role.